WS-02: Test Card TC-02 — Tethered Lift & Hover Worksheet

VERTICAL GUIDE-WIRE EQUILIBRIUM & ATTITUDE STABILIZATION AUTHORITY

AIRCRAFT: MCO-1D Mark 1	**CARD ID:** TC-02	**DATE:** ____ / ____ / 2026	**LOCATION:** _____
OPERATOR / TECH: _____	**AIRFRAME S/N:** MCO-1D-001	**AMBIENT TEMP:** _____ °C	**BARO / QNH:** _____ hPa

1. Vertical Force Equilibrium Discovery (Weightless Unstick Point)

Aircraft suspended on low-friction 4mm vertical steel guide cable. Advance frequency slowly in 0.05 Hz increments until aircraft lifts off ground stand and floats weightlessly (T/W = 1.00).

Equilibrium Parameter	Expected Range	Actual Measured Value	Engineering Notes / Telemetry Source
Weightless Unstick Frequency ($f_{ ext{hover}}$)	*1.85 to 2.10 Hz*	[ ] Hz	Frequency where load cell reaches 0.0 N tare
Commanded Flapping Amplitude	*± 75.0° (150° total stroke)*	[ ] °	Stroke angle setting on ODrive controllers
Total 6S Bus Current at Steady Hover	*18.0 to 24.0 A*	[ ] A	Summed current across all 8 ODrive inverters
Total Electrical Power at Steady Hover	*400 to 530 W*	[ ] W	$P_{ ext{elec}} = V_{ ext{bus}} imes I_{ ext{bus}}$ (Target: ~475 W)
Guide Wire Tare Collar Friction	*< 2.0 N friction*	[ ] N	Measured by lifting aircraft manually with fish scale
LiDAR Height at Steady Hover Hold	*1.0 to 1.5 m AGL*	[ ] m	TF-Luna ground altitude reading

2. Attitude Stabilization & Control Authority (Step Stick Injections)

While suspended in steady hover trim, inject ±5° stick pulses from RC transmitter / Ground Station. Verify angular response rates from Pixhawk gyro telemetry.

Control Channel	Commanded Injection	Min Required Rate	Measured Rate	Response Character / Damping
Pitch Up (AoA Fore/Aft Diff)	+ 5.0° Elevator Stick	*≥ 15.0 °/s*	[ ] °/s	[ ] Deadbeat [ ] Light Osc [ ] Sluggish
Pitch Down (AoA Fore/Aft Diff)	- 5.0° Elevator Stick	*≥ 15.0 °/s*	[ ] °/s	[ ] Deadbeat [ ] Light Osc [ ] Sluggish
Roll Right (Port/Stbd Amp Diff)	+ 5.0° Aileron Stick	*≥ 20.0 °/s*	[ ] °/s	[ ] Deadbeat [ ] Light Osc [ ] Sluggish
Roll Left (Port/Stbd Amp Diff)	- 5.0° Aileron Stick	*≥ 20.0 °/s*	[ ] °/s	[ ] Deadbeat [ ] Light Osc [ ] Sluggish
Yaw Right (Fore/Aft Sweep Diff)	+ 10.0° Rudder Stick	*≥ 12.0 °/s*	[ ] °/s	[ ] Deadbeat [ ] Light Osc [ ] Sluggish
Yaw Left (Fore/Aft Sweep Diff)	- 10.0° Rudder Stick	*≥ 12.0 °/s*	[ ] °/s	[ ] Deadbeat [ ] Light Osc [ ] Sluggish

GATE CERTIFICATION: TC-02 TETHERED LIFT & HOVER QUALIFICATION	**STATUS: [ ] PASS [ ] CONDITIONAL [ ] FAIL / RE-TEST**
INSPECTOR / LEAD: _____	**SIGNATURE & DATE:** _____
REMARKS / DISCREPANCIES: _____	